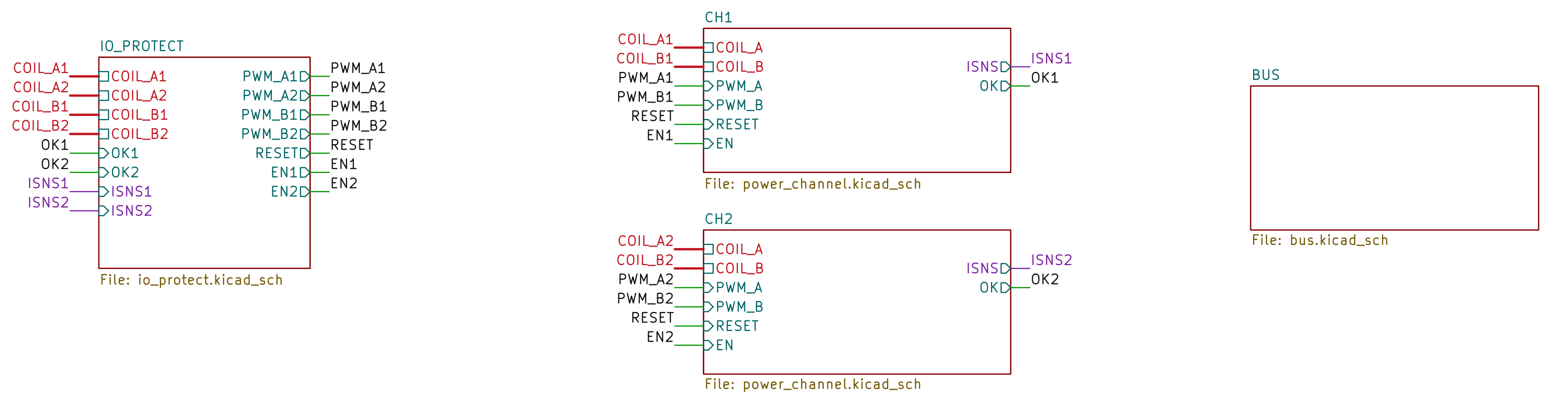
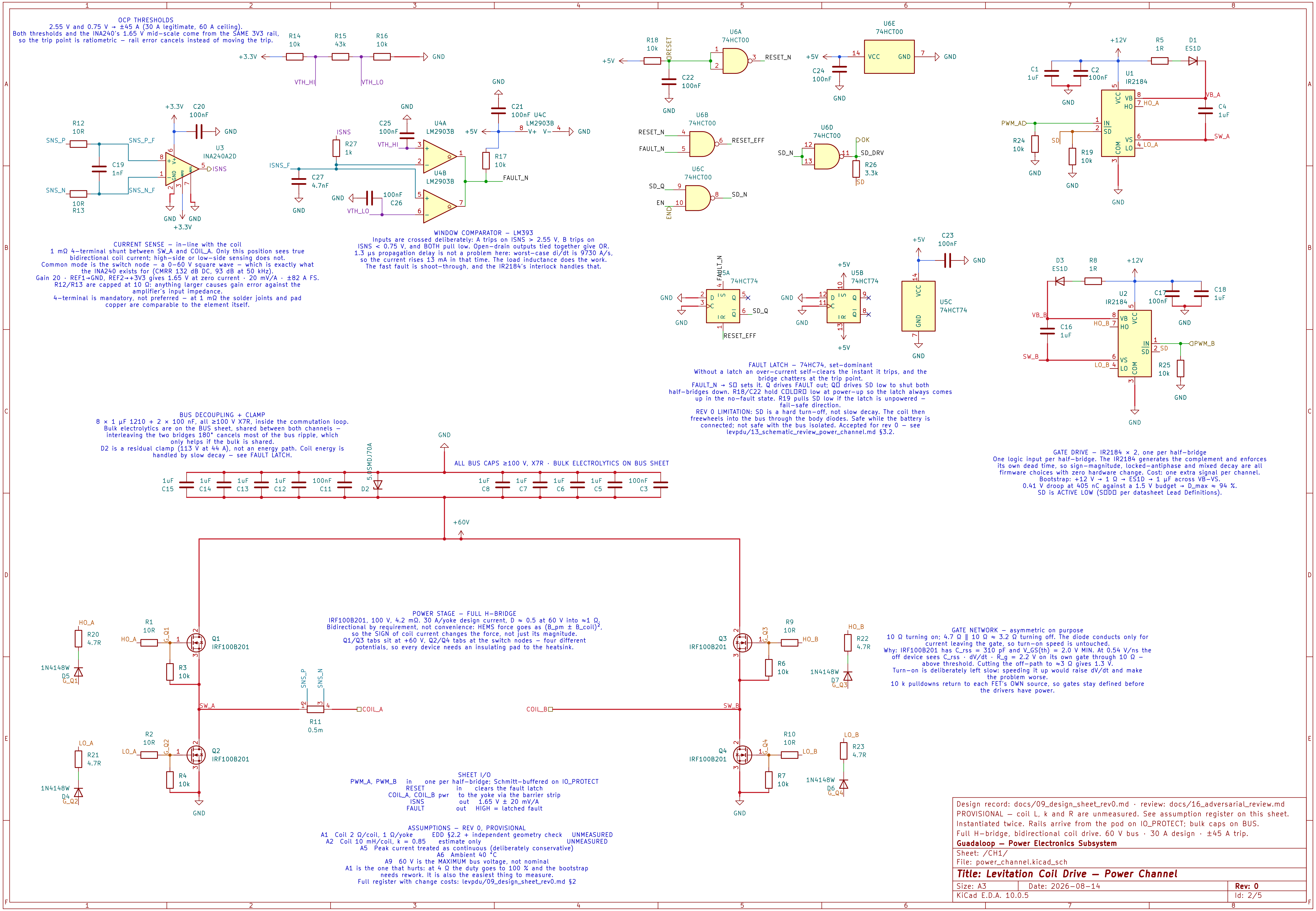


Sheet: /
File: final_lev.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 10.0.5 | Id: 1/5



OCP THRESHOLDS
2.55 V and 0.75 V → ±45 A (30 A legitimate, 60 A ceiling).
Both thresholds and the INA240's 1.65 V mid-scale come from the SAME 3V3 rail,
so the trip point is ratiometric – rail error cancels instead of moving the trip.

CURRENT SENSE – in-line with the coil
1 mΩ 4-terminal shunt between SW_A and COIL_A. Only this position sees true
bidirectional coil current; high-side or low-side sensing does not.
Common mode is the switch node – a 0–60 V square wave – which is exactly what
the INA240 exists for (CMRR 132 dB DC, 93 dB at 50 kHz).
Gain 20 · REF1→GND, REF2→+3V3 gives 1.65 V at zero current · 20 mV/A · ±82 A FS.
R12/R13 are capped at 10 Ω: anything larger causes gain error against the
amplifier's input impedance.
4-terminal is mandatory, not preferred – at 1 mΩ the solder joints and pad
copper are comparable to the element itself.

WINDOW COMPARATOR – LM393
Inputs are crossed deliberately: A trips on ISNS > 2.55 V, B trips on
ISNS < 0.75 V, and BOTH pull low. Open-drain outputs tied together give OR.
1.3 µs propagation delay is not a problem here: worst-case di/dt is 9730 A/s,
so the current rises 13 mA in that time. The load inductance does the work.
The fast fault is shoot-through, and the IR2184's interlock handles that.

FAULT LATCH – 74HC74, set-dominant
Without a latch an over-current self-clears the instant it trips, and the
bridge chatters at the trip point.
FAULT_N → S□ sets it. Q drives FAULT out; Q□ drives SD low to shut both
half-bridges down. R18/C22 hold C□L□R□ low at power-up so the latch always comes
up in the no-fault state. R19 pulls SD low if the latch is unpowered –
fail-safe direction.
REV 0 LIMITATION: SD is a hard turn-off, not slow decay. The coil then
freewheels into the bus through the body diodes. Safe while the battery is
connected; not safe with the bus isolated. Accepted for rev 0 – see
levpdu/13_schematic_review_power_channel.md §3.2.

BUS DECOUPLING + CLAMP
8 × 1 µF 1210 + 2 × 100 nF, all ≥100 V X7R, inside the commutation loop.
Bulk electrolytics are on the BUS sheet, shared between both channels –
interleaving the two bridges 180° cancels most of the bus ripple, which
only helps if the bulk is shared.
D2 is a residual clamp (113 V at 44 A), not an energy path. Coil energy is
handled by slow decay – see FAULT LATCH.

GATE DRIVE – IR2184 × 2, one per half-bridge
One logic input per half-bridge. The IR2184 generates the complement and enforces
its own dead time, so sign-magnitude, locked-antiphase and mixed decay are all
firmware choices with zero hardware change. Cost: one extra signal per channel.
Bootstrap: +12 V → 1 Ω → ES1D → 1 µF across VB–VS.
0.41 V droop at 405 nC against a 1.5 V budget → D_max ≈ 94 %.
SD is ACTIVE LOW (S□D□ per datasheet Lead Definitions).

POWER STAGE – FULL H-BRIDGE
IRF100B201, 100 V, 4.2 mΩ. 30 A/yoke design current, D ≈ 0.5 at 60 V into ≈1 Ω.
Bidirectional by requirement, not convenience: HEMS force goes as $(B_{pm} \pm B_{coil})^2$,
so the SIGN of coil current changes the force, not just its magnitude.
Q1/Q3 tabs sit at +60 V, Q2/Q4 tabs at the switch nodes – four different
potentials, so every device needs an insulating pad to the heatsink.

GATE NETWORK – asymmetric on purpose
10 Ω turning on; 4.7 Ω ‖ 10 Ω ≈ 3.2 Ω turning off. The diode conducts only for
current leaving the gate, so turn-on speed is untouched.
Why: IRF100B201 has C_rss = 310 pF and V_GS(th) = 2.0 V MIN. At 0.54 V/ns the
off device sees C_rss · dV/dt · R_g = 2.2 V on its own gate through 10 Ω –
above threshold. Cutting the off-path to ≈3 Ω gives 1.3 V.
Turn-on is deliberately left slow: speeding it up would raise dV/dt and make
the problem worse.
10 k pulldowns return to each FET's OWN source, so gates stay defined before
the drivers have power.

SHEET I/O

PWM_A, PWM_B	in	one per half-bridge; Schmitt-buffered on IO_PROTECT
RESET	in	clears the fault latch
COIL_A, COIL_B	pwr	to the yoke via the barrier strip
ISNS	out	1.65 V ± 20 mV/A
FAULT	out	HIGH = latched fault

ASSUMPTIONS – REV 0, PROVISIONAL

A1	Coil 2 Ω/coil, 1 Ω/yoke	EDD §2.2 + independent geometry check	UNMEASURED
A2	Coil 10 mH/coil, k = 0.85	estimate only	UNMEASURED
A5	Peak current treated as continuous (deliberately conservative)		
A6	Ambient 40 °C		
A9	60 V is the MAXIMUM bus voltage, not nominal		

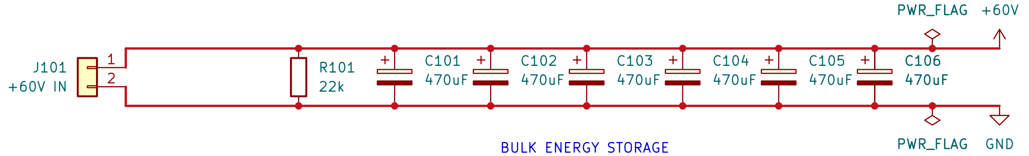
A1 is the one that hurts: at 4 Ω the duty goes to 100 % and the bootstrap
needs rework. It is also the easiest thing to measure.
Full register with change costs: levpdu/09_design_sheet_rev0.md §2

Design record: docs/09_design_sheet_rev0.md · review: docs/16_adversarial_review.md
PROVISIONAL – coil L, k and R are unmeasured. See assumption register on this sheet.
Instantiated twice. Rails arrive from the pod on IO_PROTECT; bulk caps on BUS.
Full H-bridge, bidirectional coil drive. 60 V bus · 30 A design · ±45 A trip.
Guadaloop – Power Electronics Subsystem
Sheet: /CH1/
File: power_channel.kicad_sch

Title: Levitation Coil Drive – Power Channel

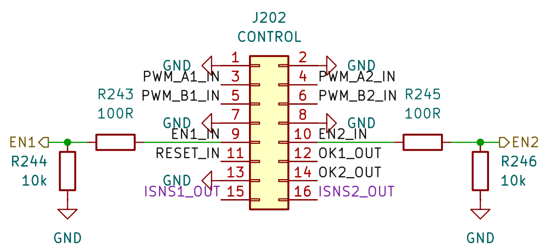
Size: A3 | Date: 2026-08-14 | **Rev: 0**
KiCad E.D.A. 10.0.5 | Id: 2/5

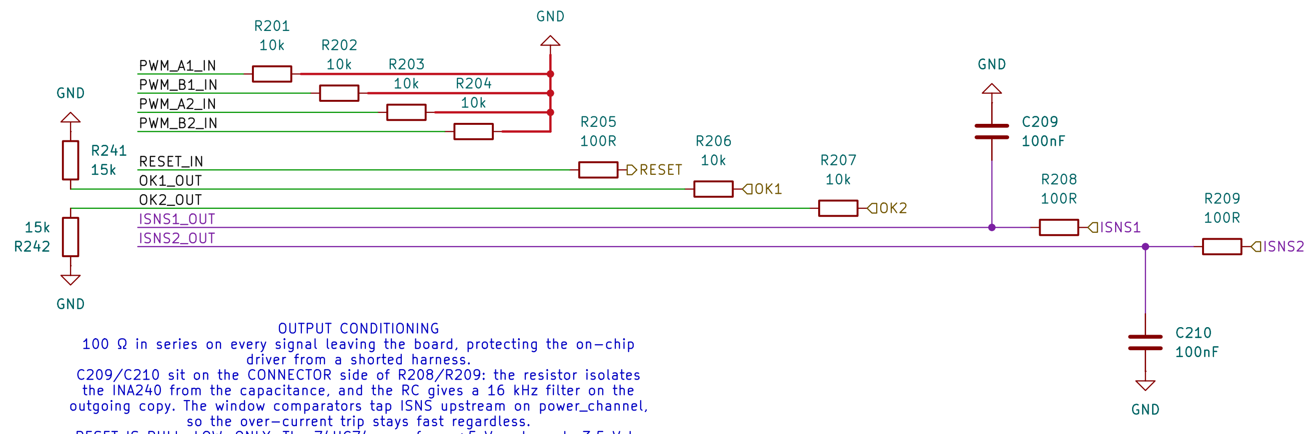
BUS FUSING — NOT ON THIS BOARD
The 60 V bus fuse is mounted in the harness at the battery:
70 A Littelfuse MIDI HP, 70 VDC rated, 2500 A interrupting.
Most automotive fuses are 32 V and are NOT acceptable here.
It protects the battery cable as well as this board. There is no
fuse between J101 and the bridges — everything downstream of J101
depends on it.



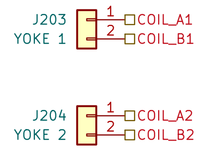
BULK ENERGY STORAGE
6 × 470 µF, shared by both power_channel instances. Sized by RIPPLE
CURRENT, not capacitance: ~20 A RMS on the shared bus once the two
channels are interleaved 180°. Select parts so the ripple ratings
sum past 20 A — the count follows the part.
R101 bleeds 5.1 J of stored energy: under 10 V in ~110 s.

Bank sized for unipolar PWM with channels interleaved 180 deg (A14). Locked antiphase needs ~52 A
Shared by both power_channel instances.
60 V battery entry, bulk energy storage, bleed. Fuse is in the harness, not on this board.
Guadaloop — Power Electronics Subsystem
Sheet: /BUS/
File: bus.kicad_sch
Title: Levitation Coil Drive — Bus / Power Entry
Size: A4 | Date: 2026-08-13 | **Rev: 0**
KiCad E.D.A. 10.0.5 | Id: 3/5

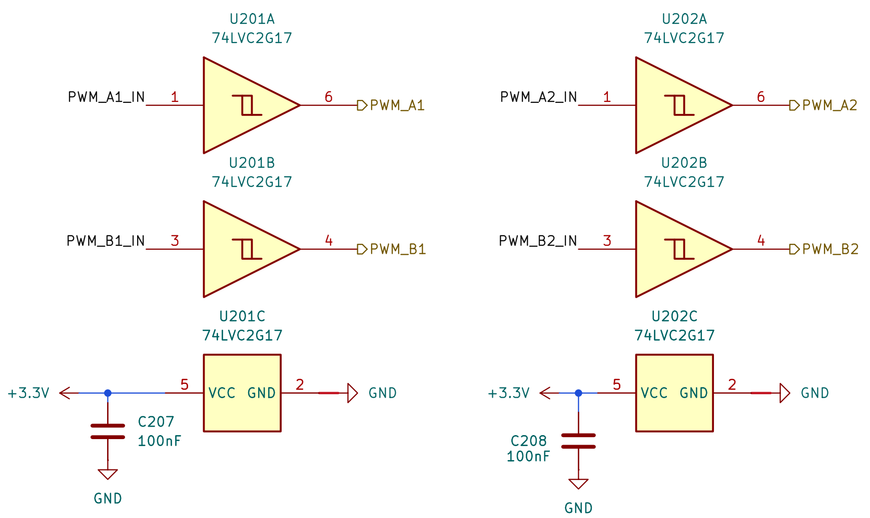


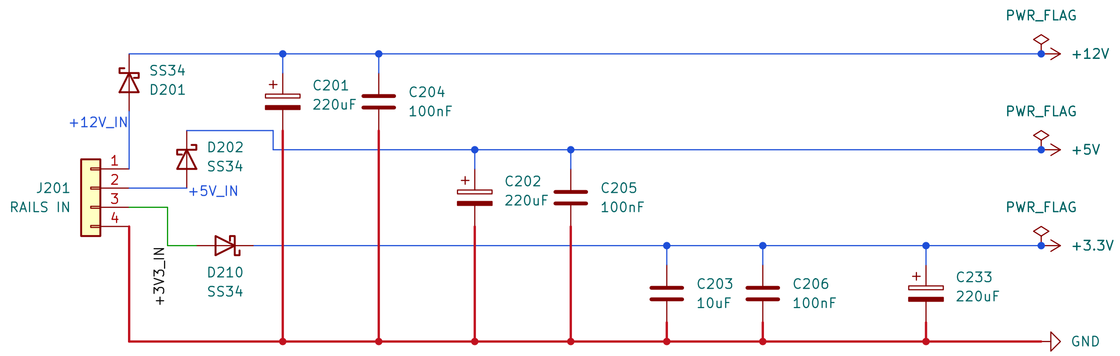


OUTPUT CONDITIONING
100 Ω in series on every signal leaving the board, protecting the on–chip driver from a shorted harness.
C209/C210 sit on the CONNECTOR side of R208/R209: the resistor isolates the INA240 from the capacitance, and the RC gives a 16 kHz filter on the outgoing copy. The window comparators tap ISNS upstream on power_channel, so the over–current trip stays fast regardless.
RESET IS PULL–LOW–ONLY. The 74HC74 runs from +5 V and needs 3.5 V to release CLR. Drive RESET low to clear a fault, then RELEASE it – the on–board pull–up provides the high level. A 3.3 V MCU driving it high would sit at 3.3 V, which is ambiguous, and you could lose fault latching entirely.



PWM INPUT BUFFERING
74LVC2G17 running from +3.3 V. Chosen because LVC inputs accept up to 5.5 V regardless of VCC – so a 3.3 V or a 5 V MCU both work, with no assumption about which. Output drives the IR2184 (VIH 2.7 V min) with 0.6 V margin; the datasheet lists 3.3 V drive as supported.
R201–R204 pull the inputs low so an unplugged harness turns both low–side FETs ON – coil shorted through Q2/Q4, slow decay. Safe state by default.





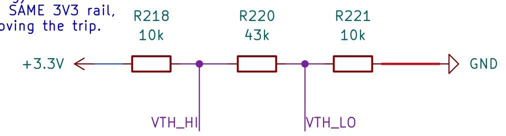
RAIL ENTRY
Rails are supplied by the pod – this board does not generate them.
D201 does two jobs: blocks reverse polarity, and isolates C201 so the 220 µF stays on this board when the pod's 12 V collapses.
C201 is the hold–up for slow–decay–on–fault: the gate drivers must stay alive ~60 ms after the 60 V bus dies. 220 µF over a 12→10.5 V window holds ~330 µC, enough for ~5 mA of driver quiescent – and only because the gates are STATIC during slow decay. If anything is still switching, gate charge dominates and this is far too small.

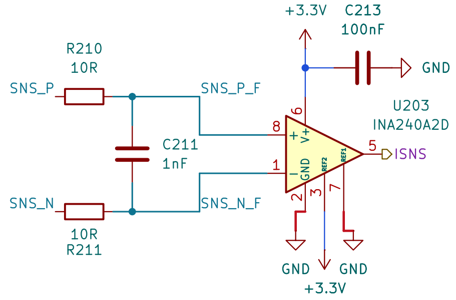
Rails are supplied by the pod – this board does not generate them.
Board I/O, PWM input buffering, rail entry and hold–up.
Guadaloop – Power Electronics Subsystem
Sheet: /IO_PROTECT/
File: io_protect.kicad_sch

Title: Levitation Coil Drive – I/O and Protection

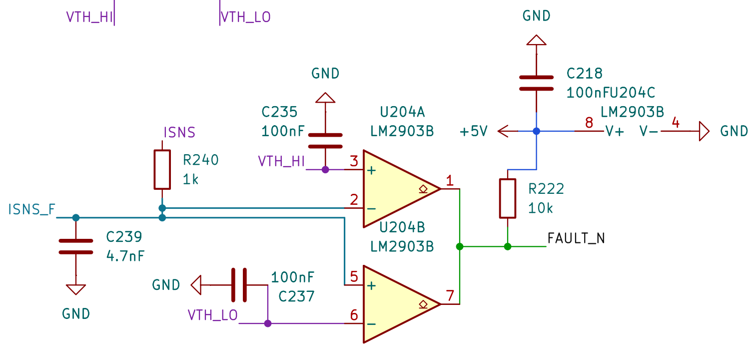
Size: A3 | Date: 2026-08-13 | **Rev: 0**
KiCad E.D.A. 10.0.5 | Id: 4/5

OCP THRESHOLDS
2.55 V and 0.75 V → ±45 A (30 A legitimate, 60 A ceiling).
Both thresholds and the INA240's 1.65 V mid-scale come from the SAME 3V3 rail,
so the trip point is ratiometric — rail error cancels instead of moving the trip.





CURRENT SENSE — in-line with the coil
1 mΩ 4-terminal shunt between SW_A and COIL_A. Only this position sees true bidirectional coil current; high-side or low-side sensing does not.
Common mode is the switch node — a 0–60 V square wave — which is exactly what the INA240 exists for (CMRR 132 dB DC, 93 dB at 50 kHz).
Gain 20 · REF1→GND, REF2→+3V3 gives 1.65 V at zero current · 20 mV/A · ±82 A FS.
R12/R13 are capped at 10 Ω: anything larger causes gain error against the amplifier's input impedance.
4-terminal is mandatory, not preferred — at 1 mΩ the solder joints and pad copper are comparable to the element itself.



WINDOW COMPARATOR — LM393
Inputs are crossed deliberately: A trips on ISNS > 2.55 V, B trips on ISNS < 0.75 V, and BOTH pull low. Open-drain outputs tied together give OR.
1.3 µs propagation delay is not a problem here: worst-case di/dt is 9730 A/s, so the current rises 13 mA in that time. The load inductance does the work.
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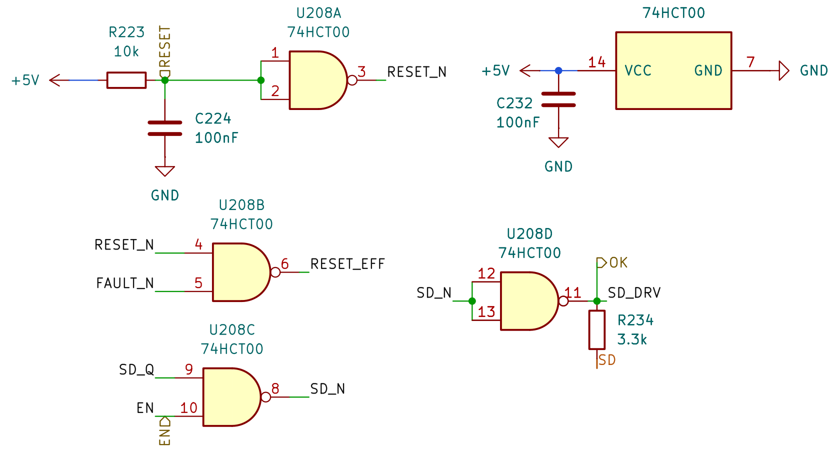
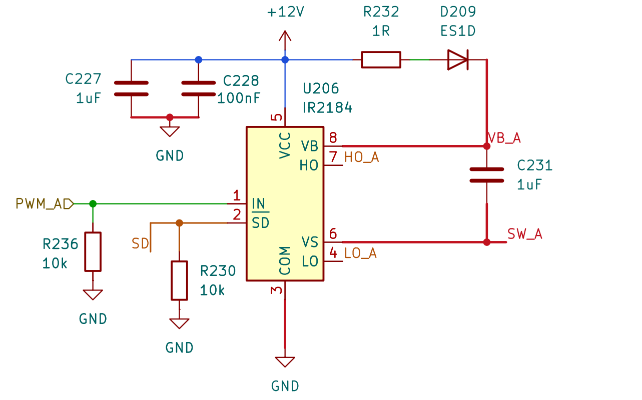
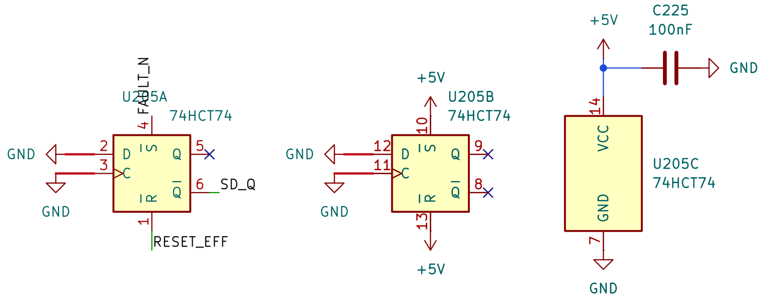
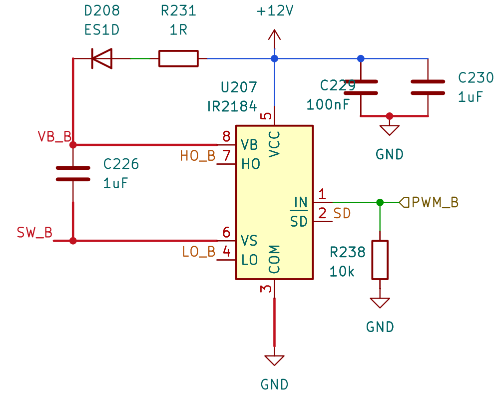






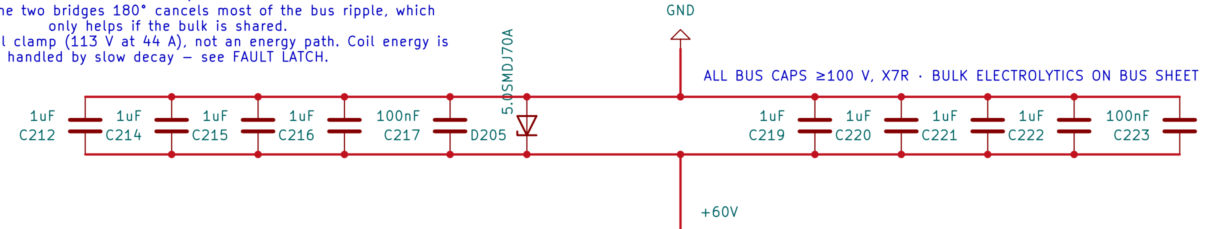
FAULT LATCH — 74HC74, set-dominant
Without a latch an over-current self-clears the instant it trips, and the bridge chatters at the trip point.
FAULT_N → S□ sets it. Q drives FAULT out; Q□ drives SD low to shut both half-bridges down. R18/C22 hold C□L□R□ low at power-up so the latch always comes up in the no-fault state. R19 pulls SD low if the latch is unpowered — fail-safe direction.
REV 0 LIMITATION: SD is a hard turn-off, not slow decay. The coil then freewheels into the bus through the body diodes. Safe while the battery is connected; not safe with the bus isolated. Accepted for rev 0 — see levpdu/13_schematic_review_power_channel.md §3.2.



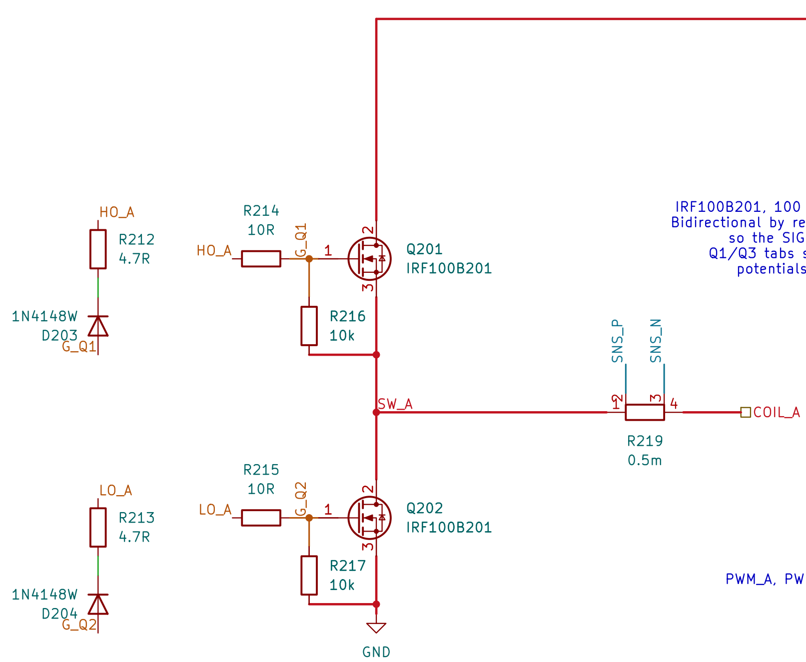
GATE DRIVE — IR2184 × 2, one per half-bridge
One logic input per half-bridge. The IR2184 generates the complement and enforces its own dead time, so sign-magnitude, locked-antiphase and mixed decay are all firmware choices with zero hardware change. Cost: one extra signal per channel.
Bootstrap: +12 V → 1 Ω → ES1D → 1 µF across VB–VS.
0.41 V droop at 405 nC against a 1.5 V budget → D_max ≈ 94 %.
SD is ACTIVE LOW (S□□□ per datasheet Lead Definitions).

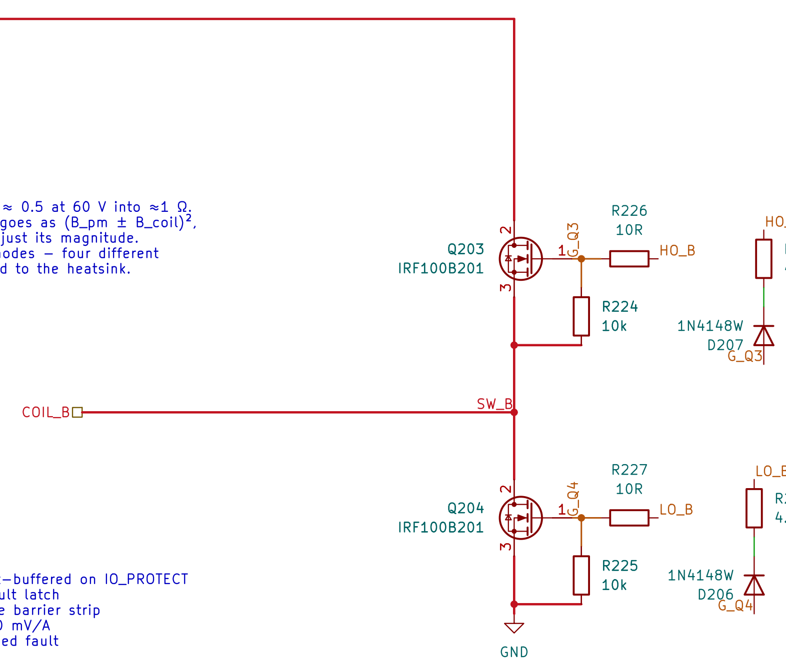
BUS DECOUPLING + CLAMP
8 × 1 µF 1210 + 2 × 100 nF, all ≥100 V X7R, inside the commutation loop.
Bulk electrolytics are on the BUS sheet, shared between both channels — interleaving the two bridges 180° cancels most of the bus ripple, which only helps if the bulk is shared.
D2 is a residual clamp (113 V at 44 A), not an energy path. Coil energy is handled by slow decay — see FAULT LATCH.

ALL BUS CAPS ≥100 V, X7R · BULK ELECTROLYTICS ON BUS SHEET



POWER STAGE — FULL H-BRIDGE
IRF100B201, 100 V, 4.2 mΩ. 30 A/yoke design current, D ≈ 0.5 at 60 V into ≈1 Ω.
Bidirectional by requirement, not convenience: HEMS force goes as (B_pm ± B_coil)², so the SIGN of coil current changes the force, not just its magnitude.
Q1/Q3 tabs sit at +60 V, Q2/Q4 tabs at the switch nodes — four different potentials, so every device needs an insulating pad to the heatsink.





GATE NETWORK — asymmetric on purpose
10 Ω turning on; 4.7 Ω ∥ 10 Ω ≈ 3.2 Ω turning off. The diode conducts only for current leaving the gate, so turn–on speed is untouched.
Why: IRF100B201 has C_rss = 310 pF and V_GS(th) = 2.0 V MIN. At 0.54 V/ns the off device sees C_rss · dV/dt · R_g = 2.2 V on its own gate through 10 Ω — above threshold. Cutting the off–path to ≈3 Ω gives 1.3 V.
Turn–on is deliberately left slow: speeding it up would raise dV/dt and make the problem worse.
10 k pulldowns return to each FET's OWN source, so gates stay defined before the drivers have power.

SHEET I/O

PWM_A, PWM_B	in	one per half–bridge; Schmitt–buffered on IO_PROTECT
RESET	in	clears the fault latch
COIL_A, COIL_B	pwr	to the yoke via the barrier strip
ISNS	out	1.65 V ± 20 mV/A
FAULT	out	HIGH = latched fault

ASSUMPTIONS — REV 0, PROVISIONAL
A1 Coil 2 Ω/coil, 1 Ω/yoke EDD §2.2 + independent geometry check UNMEASURED
A2 Coil 10 mH/coil, k = 0.85 estimate only UNMEASURED
A5 Peak current treated as continuous (deliberately conservative)
A6 Ambient 40 °C
A9 60 V is the MAXIMUM bus voltage, not nominal
A1 is the one that hurts: at 4 Ω the duty goes to 100 % and the bootstrap needs rework. It is also the easiest thing to measure.
Full register with change costs: levpdu/09_design_sheet_rev0.md §2

Design record: docs/09_design_sheet_rev0.md · review: docs/16_adversarial_review.md
PROVISIONAL — coil L, k and R are unmeasured. See assumption register on this sheet.
Instantiated twice. Rails arrive from the pod on IO_PROTECT; bulk caps on BUS.
Full H–bridge, bidirectional coil drive. 60 V bus · 30 A design · ±45 A trip.
Guadaloop — Power Electronics Subsystem
Sheet: /CH2/
File: power_channel.kicad_sch

Title: Levitation Coil Drive — Power Channel

Size: A3	Date: 2026-08-14	**Rev: 0**
KiCad E.D.A. 10.0.5		Id: 5/5